Roll No.: BL.AI.U4AID25100

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – November 2025

First Semester

Artificial Intelligence & Data Science

23AID101 Computational Thinking

Duration: 2 hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Design and Implement algorithms to solve real-world challenges.
CO02	Develop well-structured and optimized Python programs, utilizing fundamental programming concepts like data structures, control flow, and functions.
CO03	Design modular and reusable code through object-oriented programming to solve complex problems.
CO04	Implement algorithms, and develop solutions for real-world applications, including automation and data analysis, using advanced Python techniques and libraries (e.g., Pandas, NumPy).

Answer all questions.

Use proper indentation and comments for all Python code snippets. Optimally written code shall be given more consideration.

1 a. Write a Python program to print all even numbers from 1 to 10 using a for loop. [5] [CO02] [BTL 2]

b. Write a Python recursive function to calculate the factorial of a number entered by the user. [5] [CO02] [BTL 2]

2 Write a Python program to demonstrate tuple immutability by attempting to change an element and handling the error. [10][CO02] [BTL 2]

3 a. Write a Python program to input student names and marks, store them in a dictionary, and print the student with the highest marks. [6] [CO02][BTL 2]

b. Write a Python program to add a new key-value pair to a dictionary and print the updated dictionary. [4] [CO02][BTL 2]

4 a. Write a recursive Python function to calculate the power of a number (x^n) [5] [CO02] [BTL 2]

b. Describe how to define and call a function in Python. Write a function that takes two

if exponent == 0

numbers as parameters and returns their average. [5] [CO02] [BTL 2]

15 A bookstore wants to maintain its inventory using Python. The inventory includes information about the book title, author, and quantity in stock. You need to write Python programs to manage the following (Each sub question carries 2 Marks). [10][CO01][BTL 3]

a. Create a dictionary where the keys are book titles, and the values are dictionaries with keys "author" and "quantity". Write a function to add a new book to the inventory.
a. Write a function that updates the quantity of a specific book title when new stock arrives.
b. Write a program to display all books in the inventory where the quantity is below a certain threshold (e.g., less than 5).
c. Write a Python function to delete a book from the inventory if its quantity becomes zero.
d. Write a function that prints all book titles and authors sorted alphabetically by title.

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**10**	BTL 1	-
CO02	**40**	BTL 2	**40**
CO03	-	BTL 3	**10**
CO04	-	BTL 4	-

del. Books (quantity)

disp=